AI / ML • NETWORK SECURITY PROJECT

ML-Based Network Intrusion Detection

A comparative benchmark on CICIDS2017 — detection accuracy, explainability, and real-world deployability

[Ahmad Azim & Pawan Kumar] • [Computer Networks]

What Is the Project?

From static rules to learned detection

THE PROBLEM

Signature-based systems like Snort only catch attacks they already know. This project builds a machine-learning IDS that learns the statistical patterns of malicious traffic — so it can generalize to attacks it hasn't seen before.

THE APPROACH

Classify network flows

Label each traffic flow as benign or malicious using engineered statistical features.

Benchmark 5 classifiers

Compare models under identical preprocessing, splits, and metrics — a fair fight, not a single run.

Go beyond accuracy

Add explainability (SHAP) and real deployability (latency, model size) to the comparison.

2.8M

Labeled network flows analyzed (CICIDS2017)

5

ML classifiers benchmarked head-to-head

78+

Statistical flow features engineered

6

Attack categories detected: DoS, DDoS, Brute-Force, Web, Botnet, Infiltration

How It Will Be Implemented

Five stages, same data and rules for every model



Data & Preprocessing

Clean CICIDS2017, balance rare attack classes (SMOTE), select top features.

Train 5 Models

Logistic Regression, Random Forest, XGBoost, SVM, MLP — identical splits.

Evaluate Rigorously

Accuracy, F1, ROC-AUC, false-positive rate, and per-class breakdown.

Explain & Benchmark

SHAP interpretability plus latency / model-size deployability check.

Test Generalization

Validate the best model on an independent, unseen dataset.

Tools: Python • Pandas / Scikit-learn • XGBoost • imbalanced-learn (SMOTE) • SHAP • Matplotlib

What Will Be the Outcome?

Deliverables the teacher/reviewer actually sees, and what we expect to find

DELIVERABLES

Comparative benchmark report

Five classifiers ranked on accuracy, latency, and interpretability.

Feature-importance insights

SHAP shows which traffic signals drive detection decisions.

A deployment recommendation

Best model for real-time inline use vs. offline batch analysis.

A generalization verdict

Whether performance holds on data the model never trained on.

EXPECTED RESULTS (HYPOTHESES)

Aggregate accuracy
~98%+ across ensemble models (RF, XGBoost)

Rare-class recall (pre-SMOTE)
Low — shows why per-class metrics matter

Best latency / accuracy trade-off
Likely Random Forest or XGBoost

Cross-dataset accuracy
Expected drop — quantifies the generalization gap

Timeline & Key Takeaway



Weeks 1–3

Data & Features

Weeks 4–6

Model Training

Weeks 7–8

Evaluation & Explainability

Weeks 9–10

Generalization & Report

“A benchmark that doesn’t just detect intrusions — it tells you which model to trust, and why.”